

Subscription plan system overview

The SkillBridge subscription stack now offers a role-aware catalogue that ties the majority of platform features directly to plan configuration. This document explains the moving parts so product, engineering, and support teams can evolve the catalogue confidently.

Data model recap

Table	Purpose	Highlights
plans	Defines the public catalogue	Segmented by <code>target_role</code> (student or instructor), stores pricing, styling, and instructor allowances (<code>max_courses</code> , <code>ad_credits</code>).
plan_features	Feature toggles and quotas	Key/value pairs (stored as strings) that unlock modules such as ads, community, groups, tutorials, and more.
user_subscriptions	Active subscriptions	Links a user to a plan instance and records billing interval.
plan_usage_metrics	Included item usage	Tracks how many plan-covered books, classes, or tutorials a subscriber has consumed.

Tip: Feature values are normalised at runtime. Booleans can be written as `"true"` / `"false"`, numeric quotas as strings (for example `"5"`), and unlimited allowances as `"null"`.

Feature catalogue

The seed file `backend/src/seeds/09_plans_seed.js` now seeds a cohesive catalogue for both instructors and students:

- Community participation (`community_post`)
- Group creation and join limits (`groups_create` , `groups_join_limit`)
- Class authoring (`classes_create`) with course caps via `max_courses`
- Book downloads for students (`books_download`)
- Ad management controls (`ads_max_ads` , `ads_max_duration` , `ads_allow_branding` , `ads_show_analytics` , `ad_credits`)
- Tutorial creation, including plan-specific publish limits (`tutorials_create` , `tutorials_max_count`)
- Platform commissions (`commission_rate`)

When you need a new toggle, add it to the seed and gate the relevant module with `parsePlanFeatures`.

Shared helper utilities

- `backend/src/modules/plans/planContent.helper.js` normalises `included_plans` references and groups any catalogued content by plan. This is used by `plans.service` to hydrate the public API.
- `backend/src/modules/plans/plans.service.js` now attaches `included_classes` , `included_books` , and `included_tutorials` whenever plans are retrieved, so the UI can promote everything a subscriber receives at a glance.

These helpers have dedicated unit tests in

`backend/src/modules/plans/__tests__/planContent.helper.test.js` to protect against regressions.

Front-end experience

- `frontend/src/services/public/planService.js` converts media URLs for classes, tutorials, and books so plan cards can display thumbnails reliably.
- `frontend/src/components/website/sections/SubscriptionPlans.js` renders the enriched plan data, highlighting featured classes, tutorials, and books alongside the feature list.

- Localised strings for the new sections live in `frontend/public/locales/*/website.json` .

How gating works across modules

Module	Feature check
Ads (backend/src/modules/ads/ads.controller.js)	ads_max_ads, ads_max_duration, ads_allow_branding, ads_show_analytics, and ad_credits
Community (backend/src/modules/community/public/public.controller.js)	community_post
Groups (backend/src/modules/groups/groups.controller.js)	groups_create, groups_join_limit
Classes (backend/src/modules/classes/class.controller.js)	classes_create + max_courses
Tutorials (backend/src/modules/users/tutorials/tutorial.controller.js)	tutorials_create, tutorials_max_count
Books (backend/src/modules/books/book.service.js)	Subscription coverage handled via plan_usage_metrics

When you add a new premium capability, create a plan feature key, seed default values, and check it at the service layer before performing the action.

Separating instructor and student plans

`target_role` keeps the catalogue logically split:

- API clients can request `/plans?role=student` or `/plans?role=instructor` to fetch the relevant catalogue.
- The pricing UI renders student plans on the marketing site and instructor plans inside the instructor dashboard.
- Checkout flows stamp subscriptions with the plan role so billing, renewals, and analytics can segment revenue automatically.

With these foundations, the platform has a professional, extensible plan system where every feature and content perk is anchored to the subscription engine.